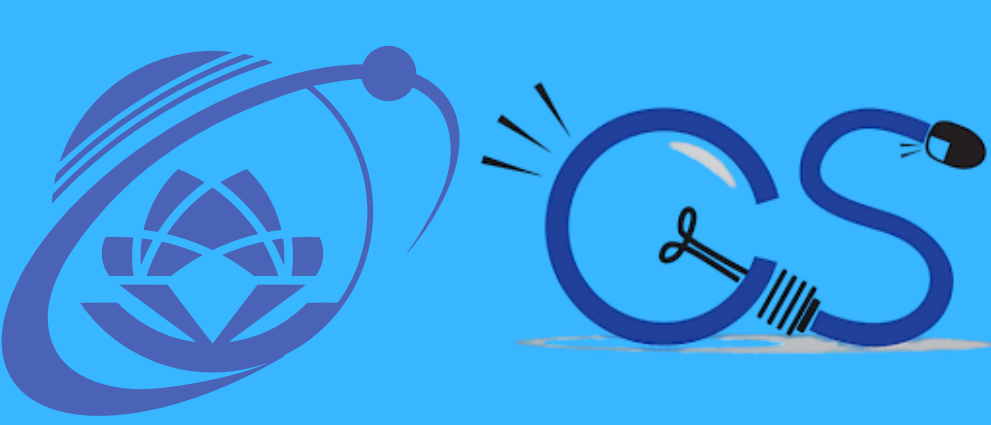




Classification of Regional Discriminatory Comments on Vietnamese Social Media

Comments

CS117.Q21.KHTN - Computational Thinking

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Introduction

Existing Vietnamese hate speech research has primarily focused on general forms of abusive language. In contrast, regional discrimination has received far less attention. To bridge this research gap, we propose a solution to detect such comments on social media, aiming to mitigate harmful content and foster a more respectful digital environment. Within the scope of this problem, regional discrimination is defined as prejudiced comments aimed at judging, degrading, excluding, stereotyping, or exhibiting unjustified favoritism towards an individual or a group of people based on their origin or place of residence.

Problem Identification

Input / Output

Input: A raw text string representing a Vietnamese user comment extracted from social media platforms (TikTok, Facebook, Reddit)

Output: The predicted label for the input text, categorized into two label types:

- Regional Discriminatory:** Prejudiced comments aimed at judging, degrading, excluding, stereotyping, or exhibiting unjustified favoritism towards targets based on their regional origin or place of residence.
- Other:** All other comments that do not meet the criteria for label Regional Discriminatory.

Scope

In-scope:

- Task:** Binary classification (Regional Discriminatory vs. Other).
- Domain:** Regional discriminatory content in Vietnam.
- Input Data:** Individual text-based comments (Facebook, TikTok, Reddit).

Out-of-scope:

- Author Identity:** Tracking or identifying comment authors.
- Explainability:** Analyzing the reasoning behind classifications.
- Other Biases:** Handling gender, race, or religious discrimination.
- Updateability:** New discriminatory terminology may be missed over time.

Assumptions

- The annotation process strictly follows the agreed definition, establishing a reliable ground truth.
- Each individual comment contains enough information for classification without requiring external context, such as the original post, previous comments, or accompanying images and videos.

Constraints

Technical Constraints:

- The input data must be plain text encoded in UTF-8.
- Multimedia components such as images, videos, or audio are not allowed.
- The length of each comment must be within the range of 3 to 512 characters.
- The input comment must be primarily written in Vietnamese. Comments written entirely in English or other foreign languages are outside the scope of the solution.

Legal Constraints:

- The input comment must not contain personally identifiable information, such as usernames, account IDs, or similar identifying information.

Requirements

Functional Requirements

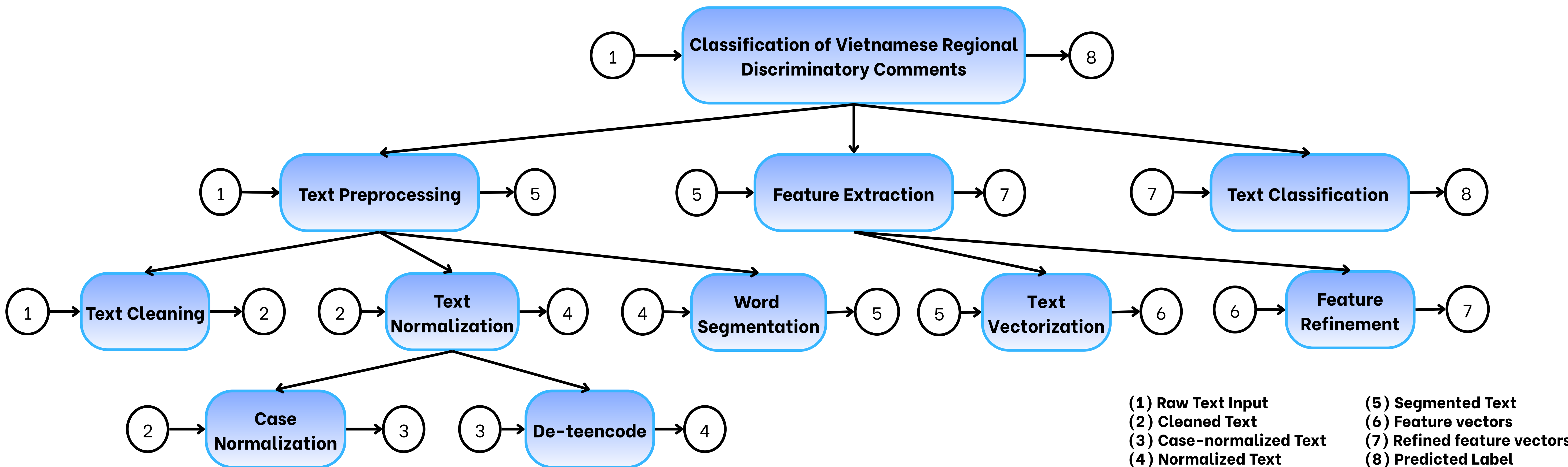
- The solution shall be able to classify an input text sequence into one of two labels based on the core definition of regional discrimination.
- The solution shall support the processing of linguistic characteristics commonly found in Vietnamese social media comments, including cases involving slang, abbreviations, and teencode.

Non-functional Requirements

- Model Performance:** Achieve an F1-score of at least 90% and an Accuracy of at least 90%.
- Processing Latency:** The total time for preprocessing and prediction shall not exceed 1 second per comment.

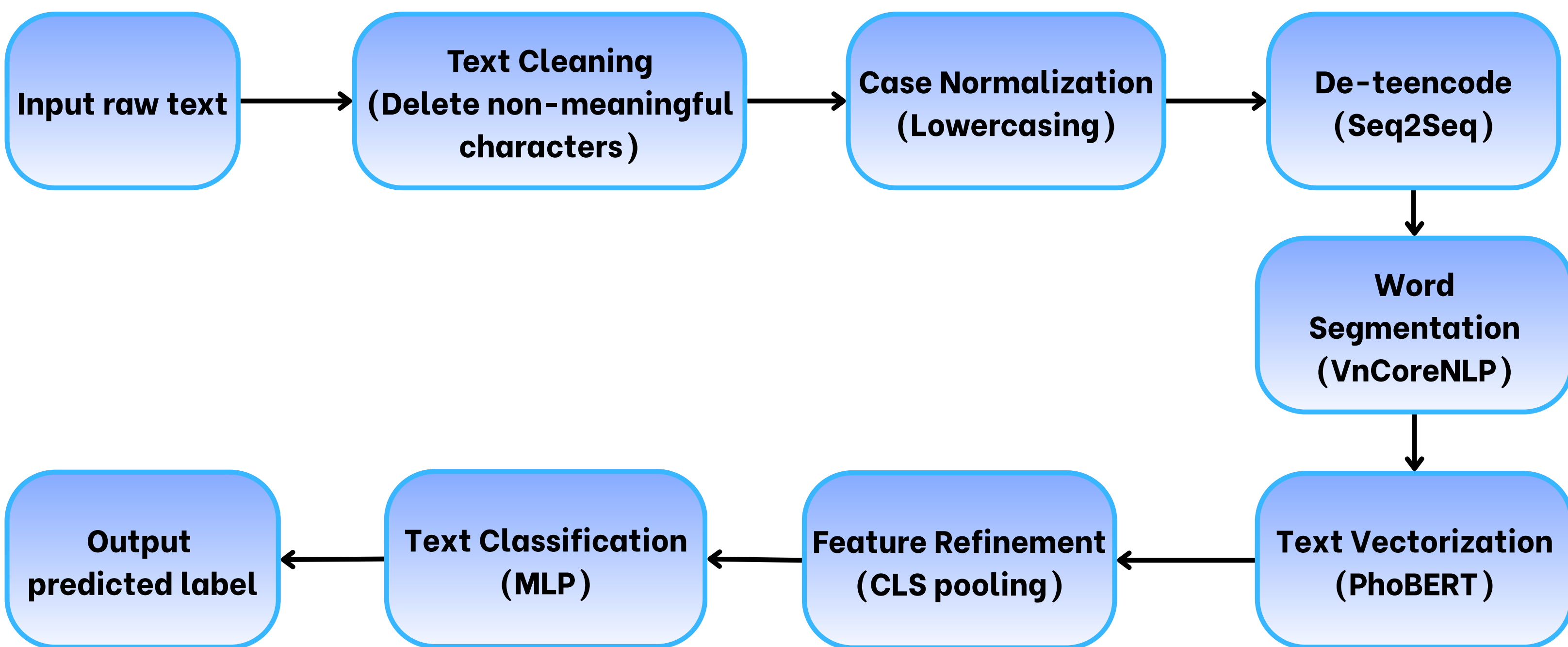
Decomposition

Breakdown Tree



Solution

Pipeline



Evaluation

Dataset

- The evaluation dataset was collected by crawling Vietnamese comments from popular social media platforms such as Facebook, TikTok, and Reddit. After collection, the data were cleaned, anonymized, and manually labeled into two classes: Regional Discriminatory and Others, based on the problem definition.
- The dataset includes both normal comments and regional discriminatory comments. In addition, the data also cover slang, abbreviations, loanwords, teencode, and distorted writing forms, in order to evaluate the solution's ability to handle real-world Vietnamese comments.

Metrics

1. Accuracy

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

3. Latency

$$\text{Latency} = t_{\text{end}} - t_{\text{start}}$$

2. F1-score

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Legend:

- TP (True Positive): The number of discriminatory comments correctly classified as discriminatory.
- FP (False Positive): The number of non-discriminatory comments incorrectly classified as discriminatory.
- FN (False Negative): The number of discriminatory comments incorrectly classified as non-discriminatory.
- t_{start} : The timestamp when receiving the input comment.
- t_{end} : The timestamp when completing processing and returning the prediction result.

Metric-Requirement Mapping Matrix & Technical Targets

Technical Requirement	Corresponding Metric	Target Value
Model Performance	Accuracy, F1-score	$\geq 90\%$
Processing Latency	Latency	< 1.0 second

Ethical & Social Issues

Ethical & Social Responsibility

- False Positives:** Neutral, humorous, sarcastic, or context-dependent comments may be misclassified.
- False Negatives:** Implicit or newly emerging discriminatory expressions may be missed.
- Bias Risk:** Social media data may reflect and reproduce existing social biases.
- Human Oversight:** The solution supports human moderators and should not automatically delete comments, suspend accounts, or impose penalties.
- Social Impact:** Helps reduce regional discrimination, but careless use may affect freedom of expression.